SABARAGAMUWA UNIVERSITY OF SRI LANKA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTING AND INFORMATION SYSTEMS
IS5103 – HIGH-PERFORMANCE COMPUTING
PRACTICAL IN-COURSE ASSESSMENT-01
DURATION- 45 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Complete all questions within the allocated examination time (45 minutes).
3. Write and execute the required programs in C using OpenMP on Ubuntu. Do not submit pseudocode in place of executable programs.
4. Save a separate .c source file for each programming question or sub-question (for example, StudentID_Q1.c).
5. Compile OpenMP programs with the GCC option and verify the output.
6. Capture clear screenshots of the source code, compilation, and execution output for each question. Include execution time and speedup results wherever required.
7. Arrange screenshots in question-number order, label them clearly, and combine them into ONE PDF document.
8. Submit the single screenshot PDF and ALL .c source files through the LMS before the examination deadline.
9. Name the PDF StudentID_OpenMP_Exam.pdf and source files StudentID_Q1.c, StudentID_Q2.c, etc.
10. Ensure screenshots and submitted .c files correspond to the same final program versions. Your work must be individual and original.

Q 1 **(30 marks)**

(a) A data-processing task must store 500,000 integer values, where element i is equal to 3 × i. Write a complete C/OpenMP program that dynamically allocates the array, validates the allocation, uses four threads to compute the values with a parallel for loop, prints the first five values after parallel processing, and frees memory.

(b) Explain why printing the first five values after the parallel loop is preferable to printing every element inside the parallel loop when measuring performance.

Q 2 **(30 marks)**

(a) Write a complete C/OpenMP program that requests three threads and uses a parallel region to print a message containing each thread's ID and the total number of participating threads.

(b) Write the commands needed to check the installed GCC version and compile and execute your source file with OpenMP enabled.

(c) Describe the difference between omp_get_thread_num() and omp_get_num_threads() in the context of your program.

Q 3 **(40 marks)**

(a) Write a complete C/OpenMP program that allocates an integer array of 1,000,000 elements, computes i × i sequentially and in parallel using four threads, times both implementations using omp_get_wtime(), and prints both execution times and the calculated speedup. Include allocation-failure handling and memory cleanup.

(b) For a program with sequential time 1.20 s and parallel time 0.40 s on four threads, calculate speedup and efficiency.

(c) A student observes a slower parallel execution for a very small array. Explain two possible reasons.